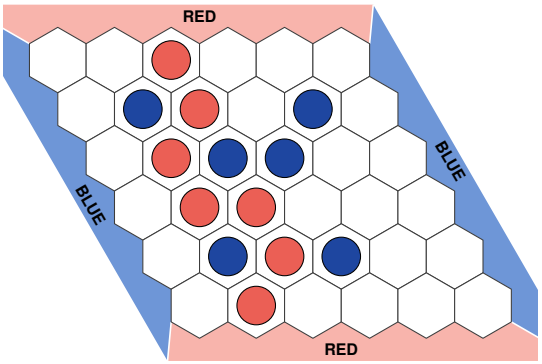
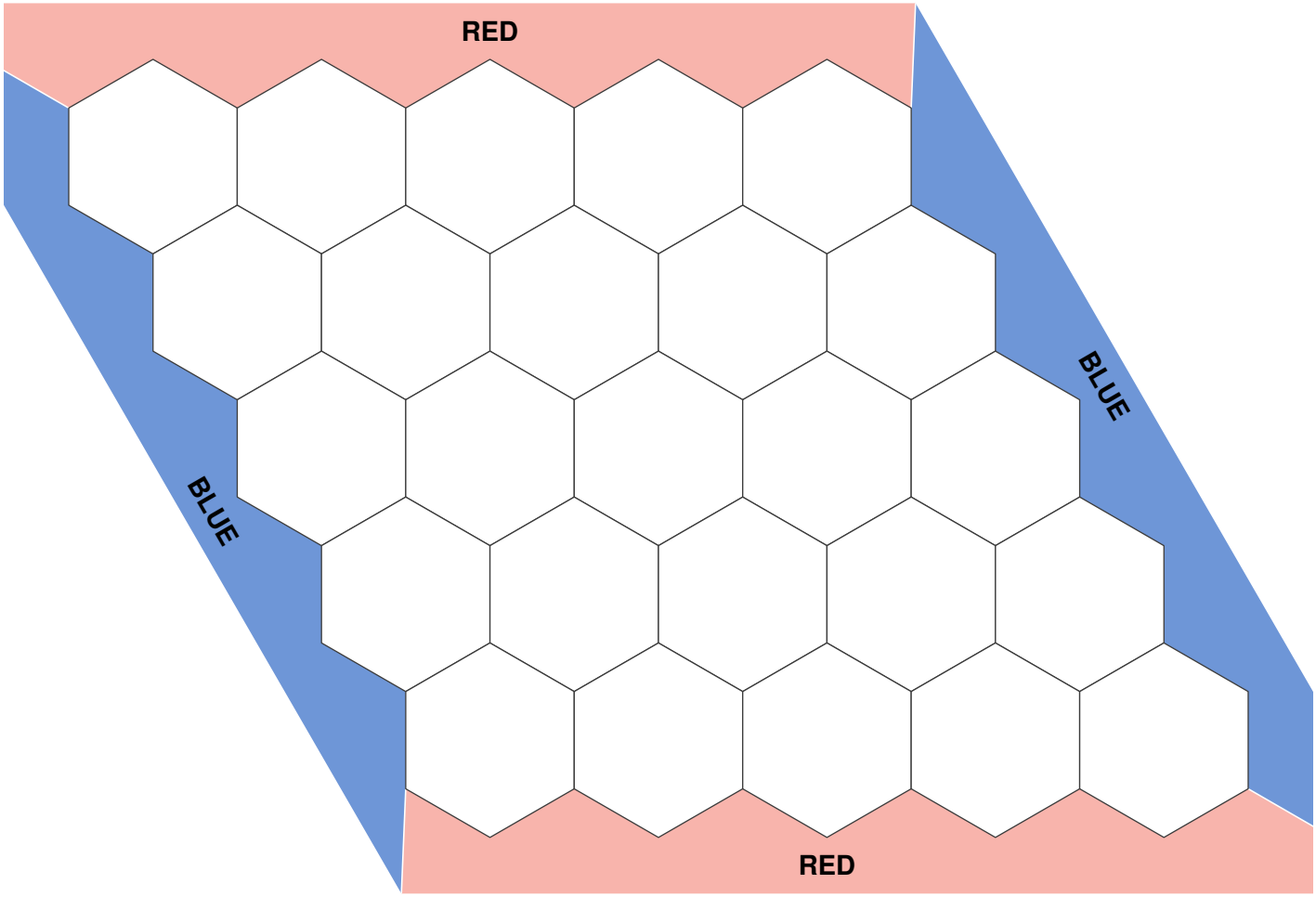
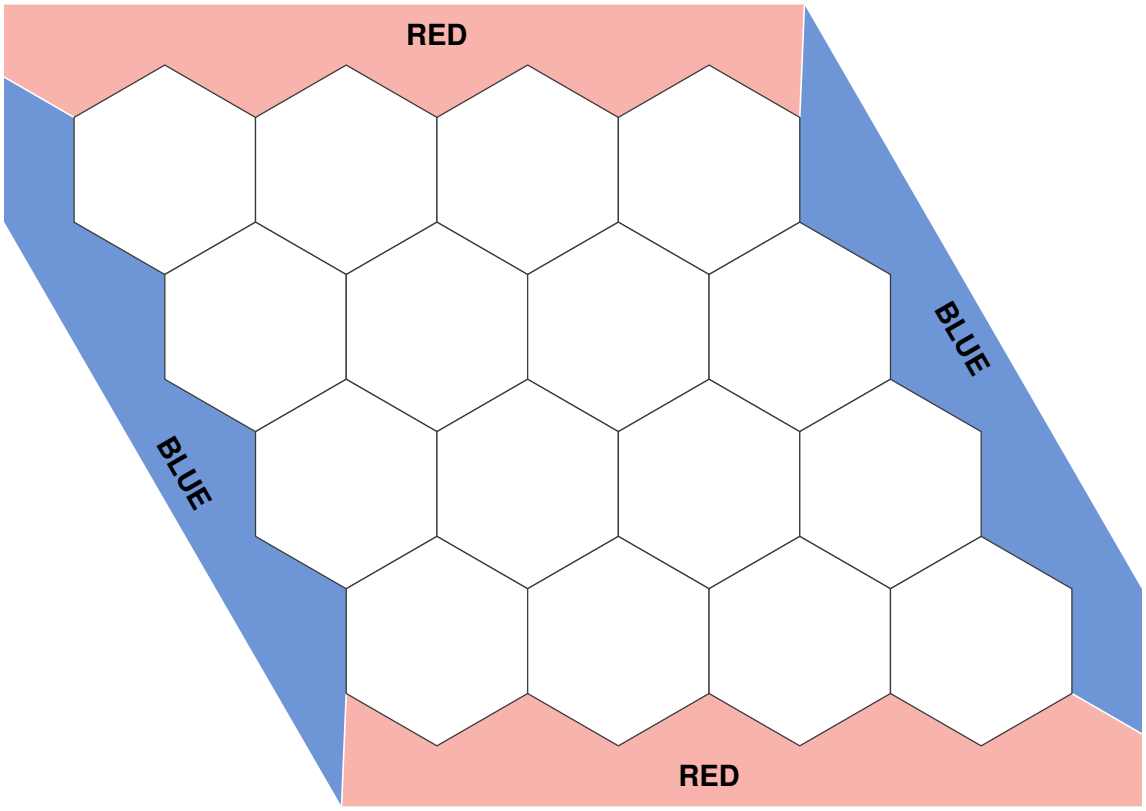
Red and Blue take turns putting a counter of their own color on an empty hexagon. Red wins by joining the two red edges with a chain of red hexagons. Blue wins by joining the two blue edges with a chain of blue hexagons. Each hexagon in a chain shares a side with the next one.



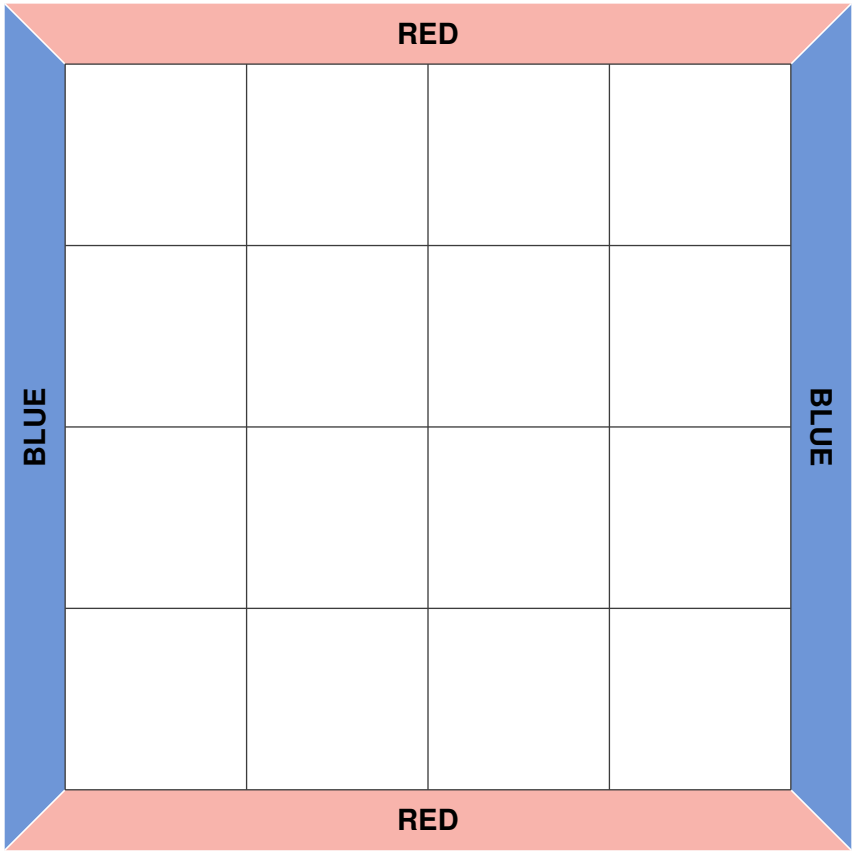
Problem 1: Play Hex on this board. Take turns going first.



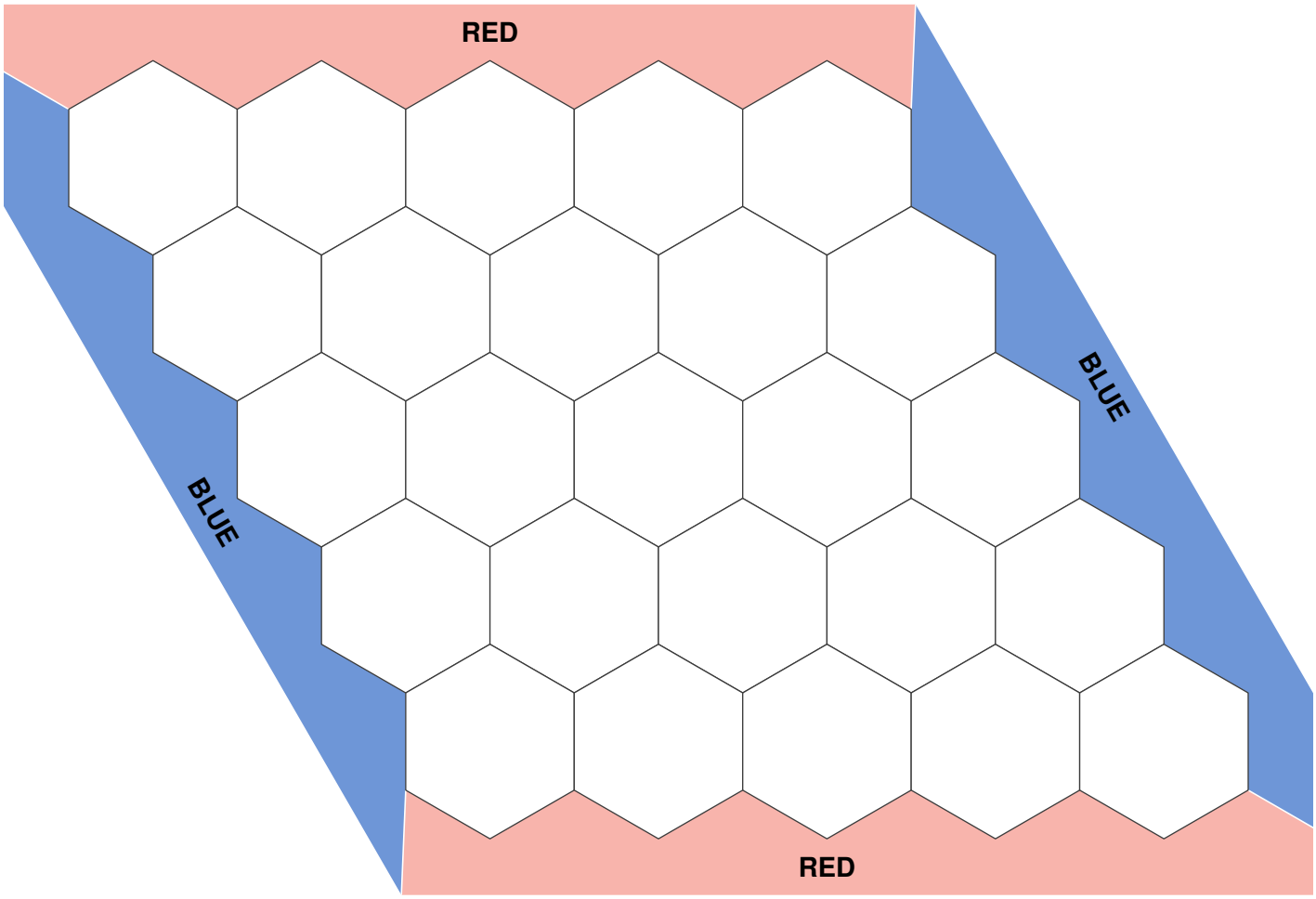
Problem 2: Can you put a red or a blue counter on every hexagon so that nobody wins? Can you do it so that Red and Blue both win?



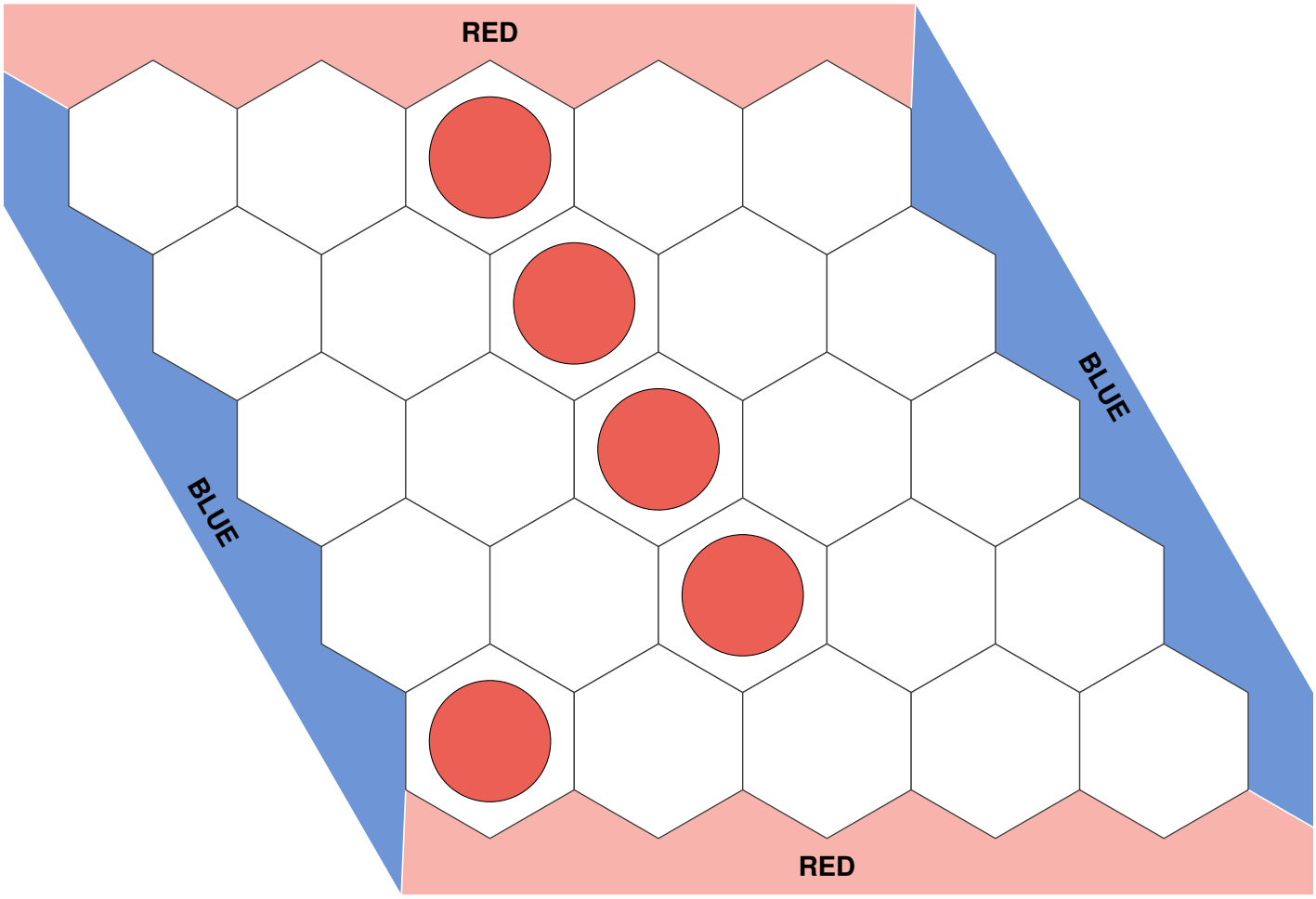
Problem 3: On this board of squares, a chain is made of squares that each share a side with the next. Put a red or a blue counter on every square so that nobody wins. Why can't you do this on the hexagon board?



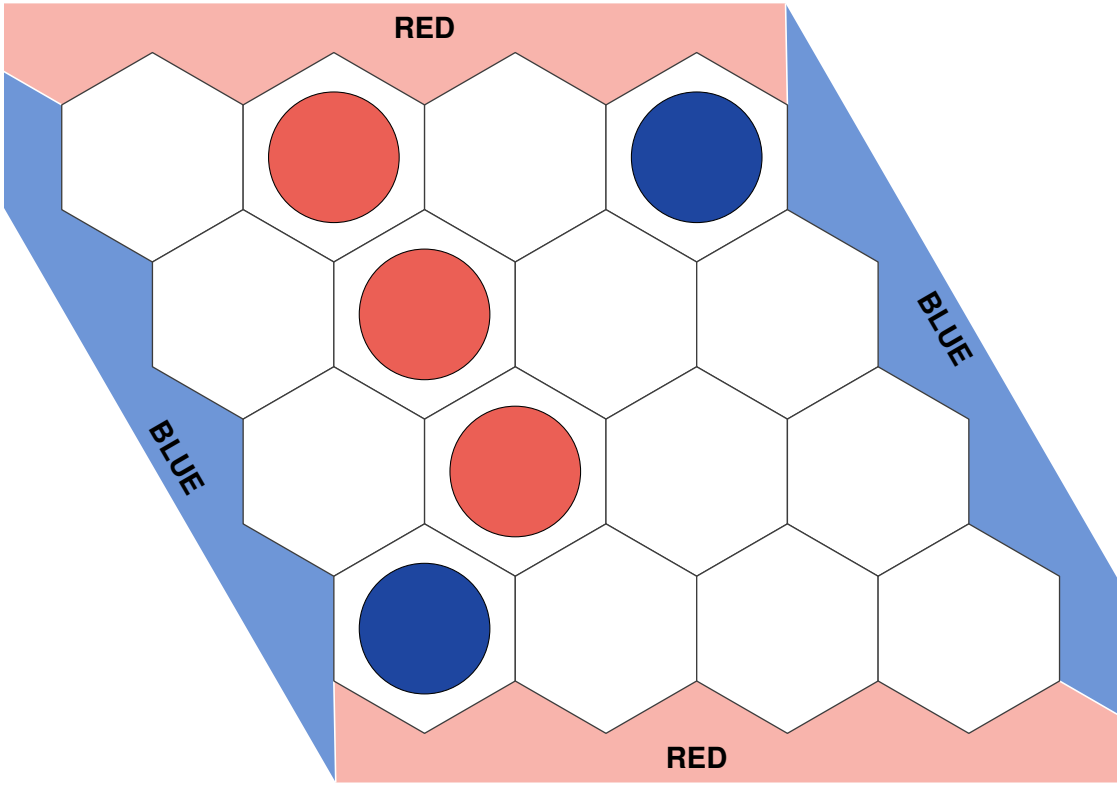
Problem 4: Put blue counters on this board so that Red cannot win, even when every other hexagon is red. Use as few blue counters as you can. Explain why fewer blue counters will not work.

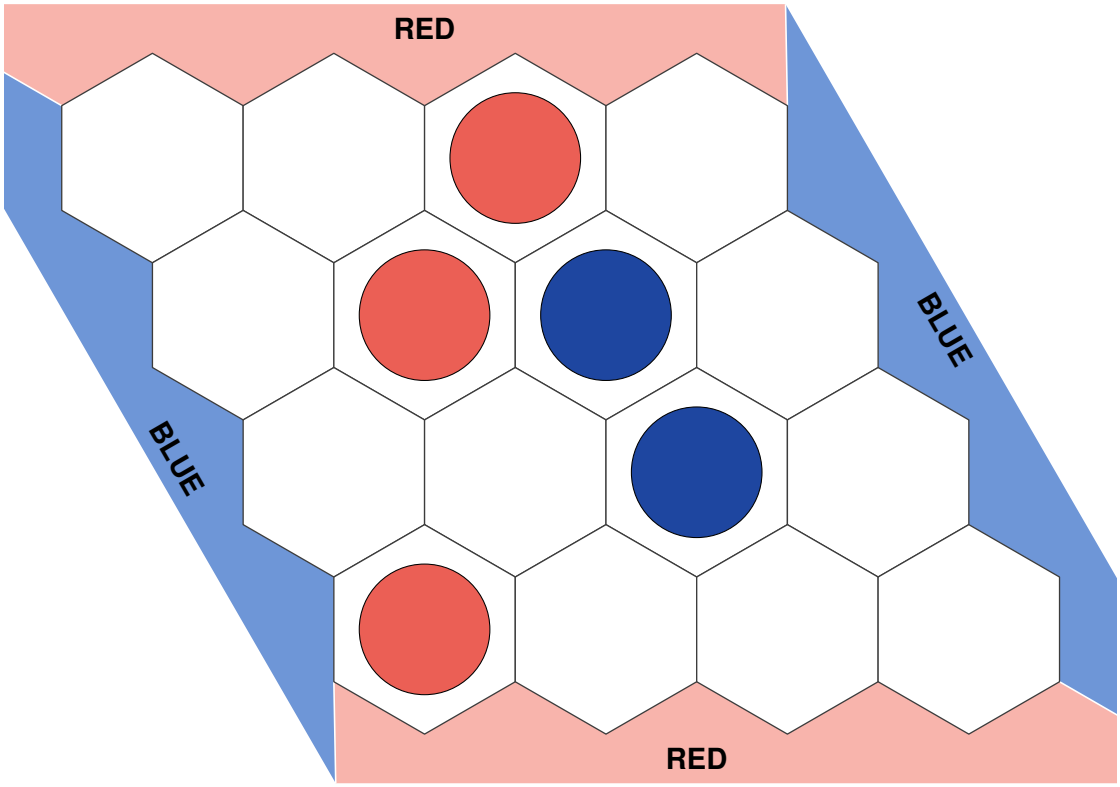


Problem 5: Red already has five counters on this board. Put blue counters on it so that Red cannot win, even when every other hexagon is red. Use as few blue counters as you can.

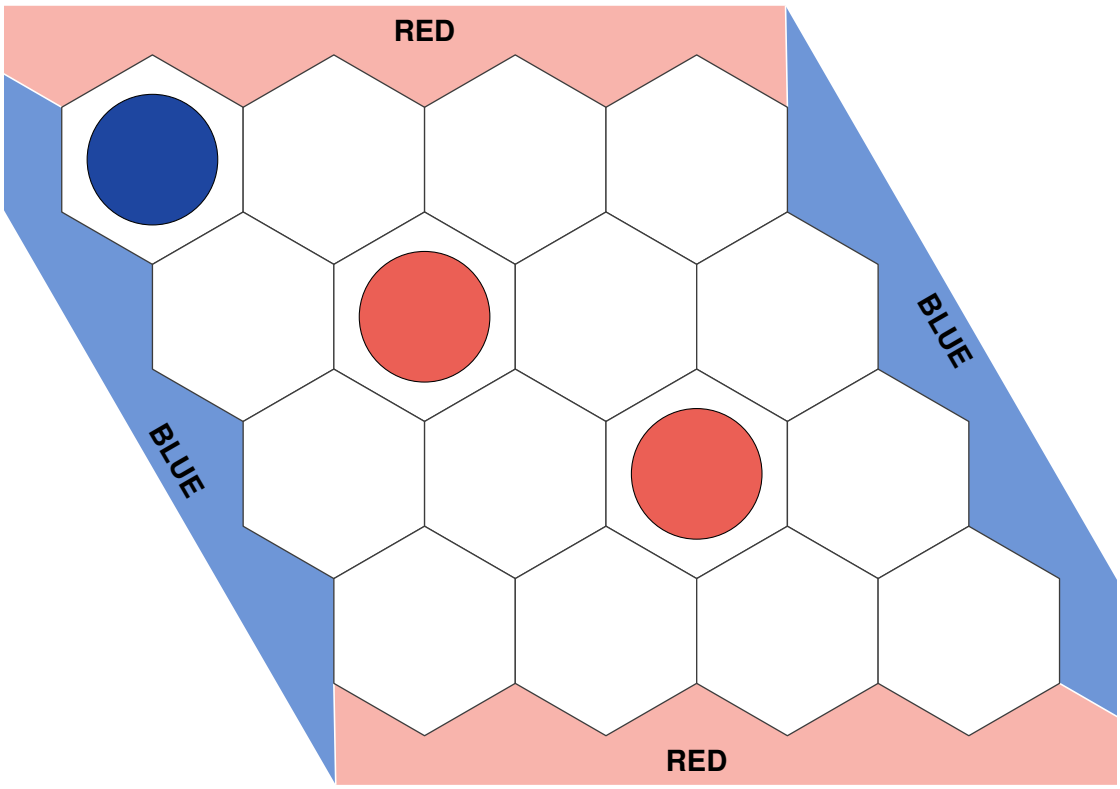


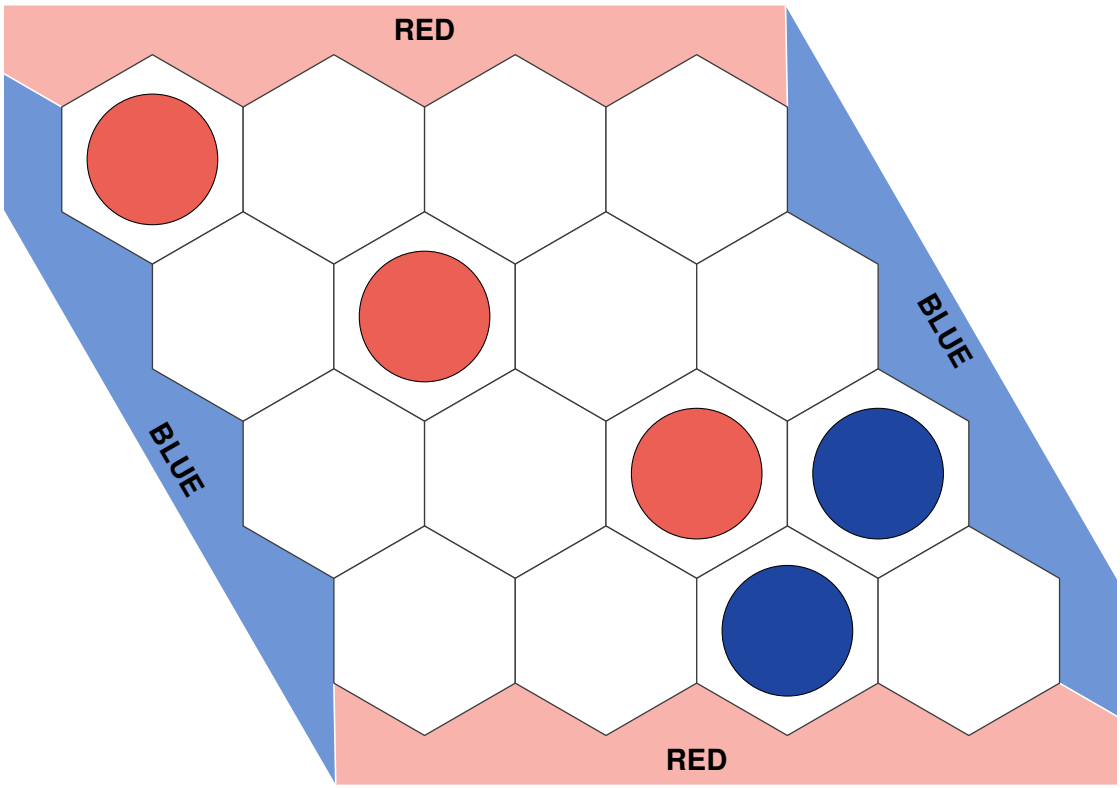
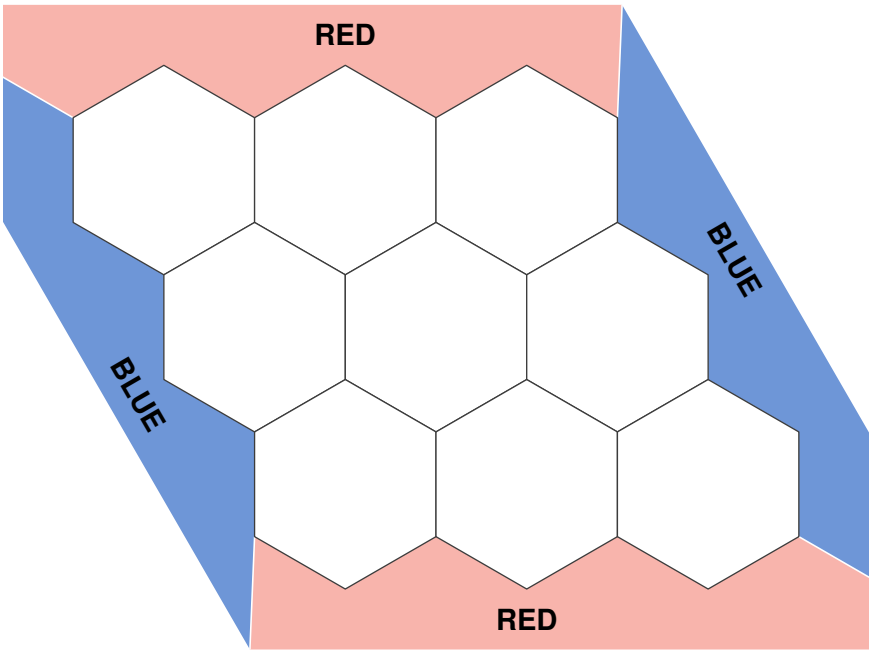
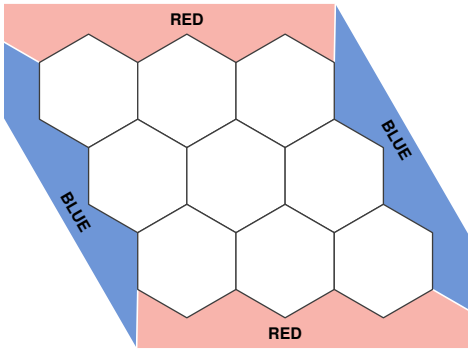
Problem 6: In each game, Red went first and now it is Blue's turn. Can Blue stop Red from winning, however Red plays? Write yes or no beside each game. If yes, mark the hexagon where Blue should play.





Problem 7: Answer the questions in Problem 6 for these two games.





Problem 8: Red goes first on this board. Find every hexagon where Red can put the first counter and then be sure to win, however Blue plays. Put an X on those hexagons on the small board.